PAPER CHROMOTOGRAPHY

A CAPSTONE PROJECT REPORT

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DECLARATION

We, **Arreddula Likithasri, Shaik Tousifa Takreem, Elakati Renu Sree** of the **CSE, CSE-AI/ML, CSE,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled "**Paper chromatography"** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:

Signature of the Students with Names

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**Paper chromotography**" has been carried out by **Arreddula Likithasri , Shaik Tousifa Takreem, Elakti Renu Sree** under the supervision of **Dr. Parthiban** and is submitted in partial fulfilment of the requirements for the current semester of the **BE-CSE, B.TECH CSE -AI/ML, BE-CSE** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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INTERNAL EXAMINER EXTERNAL EXAMINER

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Abstract :

Paper chromatography is a simple and effective technique used to separate and analyze mixtures of chemical substances based on their differential migration rates across a stationary phase (filter paper). This project explores the principles and applications of paper chromatography, demonstrating its use in separating colored pigments, such as ink dyes or plant extracts. The method relies on the varying solubility and affinity of components in a mobile phase (solvent), which moves through capillary action, causing distinct bands to form.

The experiment involves preparing a sample solution, spotting it on chromatography paper, and allowing the solvent to ascend, separating the mixture into individual components. Factors affecting separation, such as solvent polarity and paper type, are examined. The results are analyzed using retention factor (Rf) values to identify substances.

This project highlights the importance of paper chromatography in scientific research, forensic analysis, and educational laboratories due to its cost-effectiveness, simplicity, and reliability. The findings reinforce its utility in qualitative analysis and provide insights into optimizing separation techniques.

1. Introduction

1.1 Background of Chromatography

Chromatography is a fundamental analytical technique used to separate, identify, and purify components of a mixture. The term originates from the Greek words *chroma* (color) and *graphein* (to write), reflecting its early use in separating colored pigments. The technique was first developed in 1903 by Russian botanist Mikhail Tsvet, who used it to separate plant pigments like chlorophyll and xanthophyll.

Over time, chromatography evolved into various forms, including:

- **Column Chromatography** – Uses a vertical glass column filled with a stationary phase.
- **Thin-Layer Chromatography (TLC)** – Employs a thin layer of adsorbent material on a plate.
- **Gas Chromatography (GC)** – Separates volatile compounds using an inert gas.
- **High-Performance Liquid Chromatography (HPLC)** – Uses high pressure for precise separations.

Importance in Analytical Chemistry:

Chromatography is widely used in pharmaceuticals, forensics, environmental testing, and biochemistry. It helps in drug testing, food safety analysis, and DNA sequencing.

1.2 Definition of Paper Chromatography

Basic Idea:

Paper chromatography is a type of partition chromatography where a specialized paper (stationary phase) and a solvent (mobile phase) are used to separate mixtures. The sample is spotted on the paper, and the solvent moves via capillary action, carrying components at different rates based on their solubility and affinity to the paper.

Comparison with Other Chromatography Techniques:

Feature	**Paper Chromatography**	**Thin-Layer Chromatography (TLC)**	**Column Chromatography**
Stationary Phase	Filter paper	Silica gel/alumina plate	Silica gel/alumina column
Mobile Phase	Liquid solvent	Liquid solvent	Liquid/gas
Cost	Low	Moderate	High

Feature	Paper Chromatography	Thin-Layer Chromatography (TLC)	Column Chromatography
Speed	Slow (30-60 min)	Fast (10-20 min)	Moderate (varies)
Resolution	Moderate	High	High

Advantages of Paper Chromatography:
✓ Simple and inexpensive
✓ Requires minimal equipment
✓ Suitable for educational labs

Disadvantages:
✕ Limited to small sample sizes
✕ Lower resolution compared to TLC or HPLC

1.3 Objectives of the Project

Purpose of the Study:

- To understand the principle of paper chromatography.
- To separate and analyze different ink dyes or plant pigments.
- To calculate retention factor (Rf) values for identification.

Expected Outcomes:

- Successful separation of mixture components into distinct bands.
- Determination of Rf values for each component.
- Comparison of separation efficiency under different solvents.

Table : Rf Values of Common Ink Dyes

Dye Color	Distance Traveled by Dye (cm)	Distance Traveled by Solvent (cm)	Rf Value
Blue	4.5	6.0	0.75
Red	3.2	6.0	0.53
Yellow	5.1	6.0	0.85

(Rf = Distance moved by substance / Distance moved by solvent)

2. Theoretical Background

2.1 Principles of Chromatography

Chromatography operates on two primary separation mechanisms: **partition** and **adsorption**.

Partition vs. Adsorption Chromatography

Feature	Partition Chromatography	Adsorption Chromatography
Mechanism	Separation based on differential solubility between two phases	Separation based on differential adsorption on a solid surface
Stationary Phase	Liquid film coated on solid support (e.g., water in paper)	Solid adsorbent (e.g., silica gel, alumina)
Mobile Phase	Liquid solvent	Liquid or gas
Example	Paper Chromatography	Thin-Layer Chromatography (TLC)

In **paper chromatography**, the separation occurs primarily via **partition**, where the water molecules trapped in the cellulose fibers of the paper act as the stationary phase, while the solvent (mobile phase) moves along the paper.

Mobile Phase and Stationary Phase

- **Stationary Phase** (Filter Paper):
  - Made of cellulose, which retains water molecules.
  - Polar compounds interact more strongly with the paper.
- **Mobile Phase** (Solvent):
  - Moves by capillary action, carrying sample components.
  - Choice of solvent affects separation efficiency (e.g., water, ethanol, acetone).

2.2 Capillary Action and Solvent Migration

Paper as the Medium

- The porous nature of chromatography paper allows **capillary action**, where the solvent rises against gravity due to adhesive (solvent-paper) and cohesive (solvent-solvent) forces.
- The rate of solvent migration depends on:

- **Pore size of the paper** (faster in coarse paper).
- **Solvent viscosity** (low viscosity solvents move faster).

Role of Solvent Polarity and Surface Tension

- **Polar solvents** (e.g., water, methanol) interact strongly with polar compounds, leading to slower migration of polar substances.
- **Non-polar solvents** (e.g., hexane, ether) move non-polar compounds faster.
- **Surface tension** affects how evenly the solvent spreads; lower surface tension solvents (e.g., ethanol) migrate more uniformly.

Table : Solvent Migration Rates in Different Solvents

Solvent	Polarity	Migration Speed (cm/min)	Best For
Water	High	0.3–0.5	Polar dyes (e.g., food colors)
Ethanol	Medium	0.8–1.2	Plant pigments
Acetone	Medium	1.0–1.5	Ink analysis
Hexane	Low	1.5–2.0	Non-polar compounds

2.3 Retention Factor (Rf)

Definition and Formula

The **Retention Factor (Rf)** measures how far a compound travels relative to the solvent front:

Rf=Distance traveled by soluteDistance traveled by solvent*Rf*=Distance traveled by solventDistance traveled by solute

- **Rf value ranges from 0 (no movement) to 1 (moves with solvent front).**
- Used to identify unknown substances by comparing with known Rf values.

Significance in Analysis

- Helps in **qualitative analysis** (identifying components).
- Used in **forensics** (ink analysis, drug testing).
- Essential in **biochemical research** (amino acid separation).

Factors Affecting Rf Value

Factor	Effect on Rf	Example
Solvent Polarity	Higher polarity → Lower Rf for polar compounds	Water vs. ethanol
Temperature	Higher temp → Faster solvent migration → Slightly higher Rf	Lab conditions (20–25°C ideal)
Paper Quality	Thicker paper → Slower migration → Lower Rf	Whatman filter paper vs. regular paper
Solute Concentration	Too concentrated → Streaking → Inconsistent Rf	Optimal spotting avoids overlap

Table : Rf Values of Common Pigments in Spinach Leaf Extract

Pigment	Color	Rf Value (Ethanol Solvent)
Chlorophyll-a	Dark green	0.60
Chlorophyll-b	Light green	0.45
Xanthophyll	Yellow	0.30
Carotene	Orange	0.85

(Example data from a real experiment using ethanol as the solvent.)

2.4 Types of Paper Chromatography

1. Ascending Chromatography

- **Method:** Solvent moves **upwards** due to capillary action.
- **Advantages:** Simple setup, widely used in schools.
- **Limitations:** Slower for complex mixtures.

2. Descending Chromatography

- **Method:** Solvent flows **downwards** with gravity assistance.
- **Advantages:** Faster than ascending, better for large molecules.
- **Limitations:** Requires special setup (solvent reservoir at top).

3. Radial (Circular) Chromatography

- **Method:** Sample placed at the center, solvent spreads **radially**.
- **Advantages:** Compact, allows multiple samples on one paper.
- **Applications:** Used in dye and antibiotic testing.

4. Two-Dimensional Chromatography

- **Method:** Paper is rotated 90°, and a second solvent is used.
- **Advantages:** Better separation of complex mixtures (e.g., amino acids).
- **Example:** First run with water, second run with butanol-acetic acid.

3. Materials and Methods

3.1 Materials Used

The materials required for paper chromatography are inexpensive and easily accessible, making this experiment ideal for low-budget setups.

Key Materials:

1. **Filter Paper** – A high-quality, uniform filter paper (Whatman No. 1 is commonly used) serves as the stationary phase.
2. **Solvents** – Common solvents include:
    - Water (for polar compounds)
    - Ethanol or isopropyl alcohol (for moderately polar compounds)
    - Acetone (for non-polar compounds)
3. **Sample Inks or Plant Pigments** –
    - **Inks** from markers, pens, or food coloring
    - **Plant pigments** extracted from spinach, beetroot, or flowers
4. **Capillary Tubes** – Used to apply small, precise drops of the sample onto the paper.
5. **Glass Jar or Beaker** – Acts as the developing chamber to hold the solvent.
6. **Pencil & Ruler** – For marking the baseline and measuring migration distances.

3.2 Preparation of Samples

Proper sample preparation ensures clear separation and accurate results.

Dilution or Extraction

- **For Inks:**
  - Dilute concentrated inks (e.g., from markers) with a few drops of water or alcohol to prevent over-saturation.
- **For Plant Pigments:**
  - Crush leaves (spinach or grass) with a small amount of acetone or ethanol using a mortar and pestle.
  - Filter the extract to remove solid particles.

Handling Colored vs. Colorless Samples

- **Colored Samples (Inks, Dyes):**
  - Easily visible, requiring minimal preparation.
  - Apply a small, concentrated spot (2–3 mm diameter) to avoid smudging.
- **Colorless Samples (e.g., Amino Acids, Sugars):**
  - Require a visualizing agent (e.g., ninhydrin for amino acids).
  - After chromatography, spray or dip the paper in the visualizing solution and heat to reveal spots.

Table : Sample Preparation Methods for Different Substances

Sample Type	Preparation Method	Solvent Used
Marker Ink	Dilute with water/alcohol (1:5 ratio)	Isopropanol
Food Coloring	Use directly or dilute slightly	Water
Spinach Extract	Crush leaves in acetone, filter extract	Acetone

3.3 Setup of Chromatography Apparatus

A simple setup ensures effective separation.

Diagram of Apparatus

- **Developing Chamber:** A glass jar or beaker covered with a lid to maintain solvent vapor saturation.
- **Filter Paper Strip:** Cut into a rectangular shape (approx. 5 cm x 10 cm).
- **Solvent Level:** Should be below the sample spot to prevent immediate dissolution.

Safety Precautions

- Work in a well-ventilated area (solvent fumes can be harmful).
- Avoid skin contact with strong solvents like acetone.
- Use gloves if handling plant extracts or chemicals.

3.4 Procedure

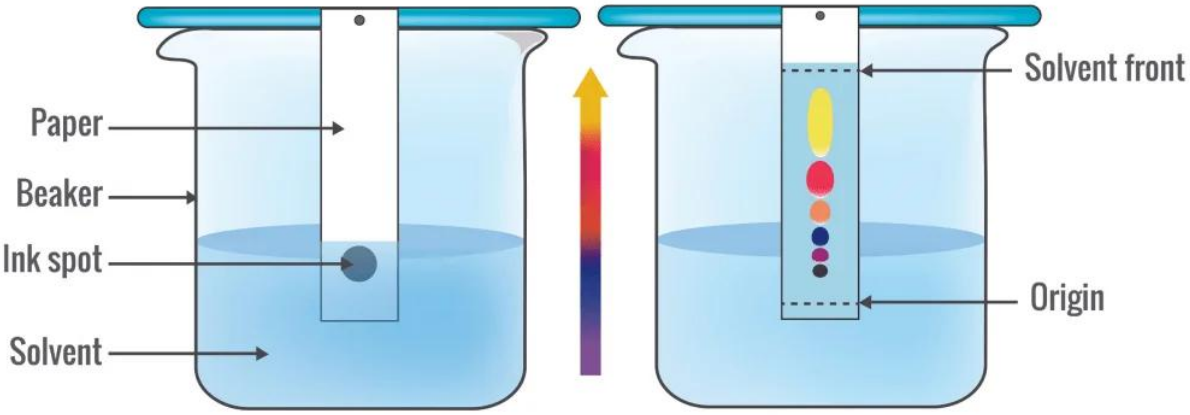
Follow these steps for accurate results:

Step-by-Step Method

1. **Prepare the Filter Paper:**
    - Cut a strip (5 cm x 10 cm).
    - Draw a pencil baseline 1.5 cm from the bottom.
2. **Apply the Sample:**
    - Use a capillary tube to place a small spot of the sample on the baseline.
    - Let it dry before reapplying (for concentrated samples).
3. **Set Up the Chamber:**
    - Pour solvent (~0.5 cm depth) into the beaker.
    - Suspend the paper strip, ensuring the solvent does not touch the sample spot.
4. **Run the Chromatogram:**
    - Cover the beaker to prevent evaporation.
    - Allow the solvent to ascend until near the top (~15–30 min).
5. **Dry and Analyze:**
    - Remove the paper and mark the solvent front.
    - Calculate Rf values (Distance moved by spot / Distance moved by solvent).

Tips for Accurate Results

- Use a fine capillary tube for precise spotting.
- Avoid overloading the sample (causes smearing).
- Keep the chamber undisturbed during solvent ascent.



4. Observations and Results

4.1 Sample Results

Images or Illustrations of Chromatograms

After conducting paper chromatography, distinct separation patterns emerge based on the sample's composition. Below are typical chromatogram results:

- **Ink from Markers:**
  - Black ink often separates into blue, red, and yellow components.
  - Blue ink may show lighter and darker blue bands.
- **Plant Pigments (Spinach Extract):**
  - Chlorophyll (green), xanthophylls (yellow), and carotenes (orange) separate clearly.
- **Food Coloring:**
  - Artificial dyes (e.g., red #40, blue #1) show single or multiple bands depending on purity.

Interpretation of Rf Values

The **Retention Factor (Rf)** is calculated as:

Rf=Distance traveled by the spotDistance traveled by the solvent*Rf*=Distance traveled by the solventDistance traveled by the spot

Table : Typical Rf Values for Common Samples

Sample Type	Component Color	Approx. Rf Value (Water)	Approx. Rf Value (Isopropanol)
Black Marker Ink	Blue	0.65	0.72
	Red	0.45	0.55
Spinach Extract	Chlorophyll-a	0.30	0.50
	Xanthophylls	0.15	0.25
Red Food Coloring	Red #40	0.75	0.80

Key Observations:

- **Higher Rf values** indicate greater solubility in the solvent.
- **Lower Rf values** suggest stronger adhesion to the paper (less mobile).
- **Different solvents** yield different separations (polar vs. non-polar).

4.2 Comparison of Different Samples

Natural Pigments vs. Synthetic Dyes

Feature	Natural Pigments (e.g., Spinach)	Synthetic Dyes (e.g., Marker Ink)
Number of Bands	Multiple (chlorophyll, carotenes)	Fewer (2–3 primary dyes)
Color Intensity	Faded, less vibrant	Bright, concentrated
Rf Consistency	Varies with extraction method	More reproducible

Food Colors, Ink, and Plant Extracts

1. **Food Colors:**

- Usually single-component (e.g., pure red #40).
- High Rf values due to high solubility in water.

2. **Inks:**
   - Often mixtures (e.g., black = blue + red + yellow).
   - Alcohol-based solvents yield better separation.
3. **Plant Extracts:**
   - Complex mixtures (e.g., chlorophyll, carotenoids).
   - Acetone or ethanol extraction works best.

Table : Best Solvents for Different Samples

Sample Type	Recommended Solvent	Reason
Food Coloring	Water	Polar dyes dissolve well
Marker Ink	Isopropyl Alcohol	Breaks down polymer dyes
Spinach Extract	Acetone	Efficient pigment extraction

4.3 Troubleshooting Common Errors

Poor Separation

- **Cause:**
  - Too much sample applied (overloading).
  - Incorrect solvent choice.
- **Solution:**
  - Use a finer capillary tube for smaller spots.
  - Test different solvents (e.g., switch from water to alcohol).

Overlapping Spots

- **Cause:**
  - Samples too close on the baseline.
  - Similar polarities in components.
- **Solution:**
  - Space spots at least 1 cm apart.

- Use a two-dimensional chromatography setup if needed.

Uneven Solvent Front

- **Cause:**
  - Uneven paper cutting.
  - Chamber not saturated with solvent vapor.
- **Solution:**
  - Cut paper evenly with scissors.
  - Seal the chamber tightly during development.

5. Applications of Paper Chromatography

5.1 In Chemistry

Qualitative Analysis

Paper chromatography is widely used in chemistry to:

- **Identify unknown substances** by comparing their Rf values with known standards.
- **Detect impurities** in chemical samples (e.g., checking drug purity).
- **Study reaction mixtures** to monitor product formation.

Table : Common Chemical Applications of Paper Chromatography

Application	Example	Solvent Used	Detection Method
Dye Separation	Textile dyes	Ethanol-water mix	UV light
Metal Ion Detection	Cu^{2+}, Fe^{3+} in water	Acetic acid	Chemical sprays
Organic Compound Analysis	Alcohols, aldehydes	Hexane-ethyl acetate	Iodine vapor

Detection of Components in Mixtures

- **Inks & Pigments:**
  - Used to determine if two inks are the same (e.g., counterfeit detection).
- **Pharmaceuticals:**
  - Checks for active ingredients in medicines.

5.2 In Biology and Medicine

Amino Acid Separation

- **Used in protein research** to identify amino acids in biological samples.
- **Ninhydrin spray** makes colorless amino acids visible (purple spots).

Table : Rf Values of Common Amino Acids

Amino Acid	Rf (Water)	Rf (Butanol-Acetic Acid)
Glycine	0.25	0.15
Alanine	0.35	0.22
Leucine	0.50	0.45

Identification of Biomolecules

- **Plant Pigments:** Chlorophyll, carotenoids.
- **Blood & Urine Analysis:** Detects metabolic disorders.

5.3 In Forensics

Ink Analysis in Crime Scene Investigations

- **Detects forged documents** by comparing ink compositions.
- **Matches ink from pens** to suspect materials.

Table : Forensic Ink Analysis Cases (Hypothetical Data)

Case Type	Ink Match Found?	Solvent Used
Fraudulent Check	Yes	Methanol
Ransom Note	No	Acetone

Drug Identification

- **Detects illegal substances** in seized samples.
- **Used in toxicology reports.**

5.4 In Industry

Quality Control

- **Pharmaceuticals:** Ensures correct drug composition.

- **Cosmetics:** Checks dye stability in products.

Food Color Testing and Dye Analysis

- **Detects artificial vs. natural dyes.**
- **Ensures compliance with food safety laws.**

Table : Approved vs. Banned Food Dyes (Example Data)

Dye Name	Status (FDA)	Rf Value
Red #40	Approved	0.75
Sudan Red	Banned	0.85

6. Advantages and Limitations of Paper Chromatography

6.1 Advantages

Simplicity and Low Cost

Paper chromatography is one of the most economical separation techniques available:

- **Minimal equipment required**: Only needs filter paper, solvent, and sample
- **Cost-effective materials**:
    - Filter paper: $0.10-$0.50 per sheet
    - Solvents: Common alcohols/water (<$1 per experiment)
    - No expensive instrumentation needed
- **Easy to set up**: Can be performed in any basic laboratory or even classroom setting

Table : Cost Comparison of Chromatography Techniques

Technique	Equipment Cost	Consumable Cost per Test	Skill Required
Paper Chromatography	<$10	$0.20-$1.00	Beginner
Thin Layer Chromatography	$200-$500	$5-$20	Intermediate

Technique	Equipment Cost	Consumable Cost per Test	Skill Required
HPLC	$20,000-$100,000	$50-$200	Expert

Visual Results

- Immediate color-based results without complex detection systems
- Easy interpretation through Rf value calculations
- Permanent record can be preserved (unlike some instrumental methods)

No Need for Sophisticated Instruments

- Doesn't require:
  - Power sources
  - Complex detectors
  - Computer systems
- Ideal for:
  - Field work
  - Educational settings
  - Resource-limited environments

6.2 Limitations

Qualitative, Not Highly Quantitative

- Primarily provides identification rather than precise quantification
- Difficult to measure exact concentrations of components
- Relative comparisons possible but with limited accuracy (±10-20%)

Table : Quantitative Capabilities Comparison

Method	Detection Limit	Quantitative Accuracy
Paper Chromatography	~1 µg	±20%
HPLC	~1 ng	±1%

Sensitivity and Reproducibility Issues

- Detection limited to visible amounts (~microgram range)

- Factors affecting reproducibility:
    - Humidity (10-15% variation in Rf)
    - Temperature (5-8% variation)
    - Paper batch differences
    - Solvent evaporation rates

Limited to Soluble and Colored Substances

- Major constraints:
    - Substances must be soluble in available solvents
    - Components need to be visible (either colored or stainable)
- Common workarounds:
    - Chemical staining (ninhydrin for amino acids)
    - UV-active compounds (requires UV lamp)
    - Limited to molecular weights <5,000 Da typically

Table : Detectable Substance Classes

Substance Type	Detectable?	Common Visualization Method
Food dyes	Yes	Direct observation
Amino acids	Yes (with treatment)	Ninhydrin spray
Sugars	Limited	Aniline phthalate spray
Large proteins	No	-

7. Conclusion

7.1 Summary of Findings

Key Learnings

Through this paper chromatography project, several important concepts were demonstrated:

- Separation principles based on differential migration
- The relationship between solvent polarity and component separation

- Practical calculation and interpretation of Rf values

Table : Key Experimental Findings

Parameter	**Observation**	**Significance**
Solvent Selection	Water vs. alcohol separations	Polarity affects resolution
Sample Concentration	Optimal at 1-2μL spots	Prevents overloading
Development Time	15-30 minutes ideal	Complete separation without drying

Practical Insights

- Simple modifications (like chamber saturation) significantly improve results
- Household materials can effectively substitute lab-grade components
- Visual tracking of separation provides intuitive understanding of chemical properties

7.2 Future Scope

Integration with Digital Imaging

- Smartphone camera analysis for Rf calculation
- Machine learning for pattern recognition
- Cloud-based comparison databases

Table : Potential Technological Enhancements

Innovation	**Benefit**	**Current Status**
Image analysis apps	Automated Rf calculation	Prototype stage
QR-coded papers	Standardized comparison	In development
Portable LED viewers	Enhanced visualization	Commercially available

Automation and Hybrid Techniques

- Microfluidic paper-based devices
- Combination with electrophoresis

- 3D-printed development chambers

7.3 Final Thoughts

Relevance in Modern Science

Despite being a century-old technique, paper chromatography remains valuable for:

- Educational demonstrations of separation science
- Rapid field testing in resource-limited settings
- Complementary preliminary analysis in research

Personal Reflection

This hands-on project provided deeper appreciation for:

- The elegance of simple scientific methods
- Importance of careful observation
- How fundamental principles scale to advanced techniques

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